

Clinical-Inspired Multi-View Retrieval with Spatial Residual Attention and Concept-Driven Explainability in Medical Images

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Abstract. Content-based image retrieval (CBIR) in medical imaging can assist clinicians by retrieving previously diagnosed cases with visual patterns similar to a query examination. However, reliable retrieval in chest radiography remains challenging because subtle pathological textures, such as early-stage lesions, are often overlooked by models that rely primarily on global representations. To address this issue, we propose a clinically inspired multi-view retrieval framework built on a modernized CNN backbone (**ConvNeXtV2**). The framework adopts a dual-branch architecture consisting of a **Canonical Branch** for capturing global anatomical context and an **Attentive Refinement Branch** with **Spatial Residual Attention (SRA)** to emphasize localized pathological cues. To improve interpretability, we introduce **Concept-to-Region Alignment for Retrieval (CRAR)**, which associates retrieved results with clinically meaningful visual concepts and highlights the corresponding image regions. Experiments on three medical imaging datasets (COVIDx, ISIC 2017, and TBX11K) demonstrate that the proposed framework consistently outperforms recent medical vision models, including DinoV2, BioMedCLIP, MedSigLIP, and MedCLIP. Additional evaluation using insertion and deletion metrics shows that our concept-driven explanations better align with clinically relevant evidence, providing more transparent and reliable retrieval results for medical image analysis.

Keywords: Medical Image Retrieval · Chest Radiography · Explainable AI · Spatial Residual Attention · Concept-to-Region Alignment · ConvNeXtV2.

1 Introduction

Medical imaging plays a critical role in modern clinical diagnosis. With the rapid growth of digital medical archives, content-based image retrieval (CBIR) has emerged as a promising decision-support tool that allows clinicians to retrieve previously diagnosed cases with visual patterns similar to a query image. By comparing a current examination with historical cases, CBIR systems can provide empirical evidence that helps reduce diagnostic uncertainty and supports clinical decision-making. For real clinical deployment, however, retrieval systems must satisfy two essential requirements: they

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must preserve subtle pathological cues and provide interpretable explanations consistent with clinical reasoning. As illustrated in Fig. 1, a clinically reliable retrieval framework should not only return visually similar cases but also provide spatially grounded explanations that allow clinicians to verify the underlying pathological evidence.

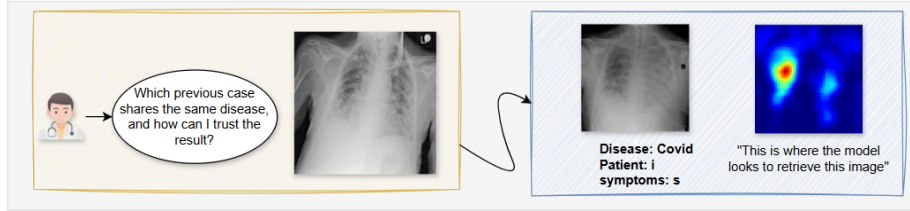


Fig. 1. Overview of the retrieval-enhanced diagnostic workflow. The system identifies pathologically similar cases and provides saliency maps for spatial grounding, enabling clinicians to verify diagnostic evidence and establish trust.

Despite recent progress in deep learning for medical imaging, two challenges remain unresolved in medical image retrieval. The first challenge is the preservation of fine-grained pathological textures. Early CBIR systems relied on handcrafted descriptors such as texture and shape statistics [9]. Deep convolutional neural networks later became the dominant paradigm for extracting visual embeddings, enabling similarity comparison through distance metrics such as cosine similarity. More recently, vision-language foundation models have been explored for medical retrieval. For example, Denner et al. [3] demonstrated that CLIP-based medical models can achieve competitive performance on radiology datasets without extensive task-specific training. However, these models are primarily optimized for cross-modal semantic alignment rather than instance-level visual similarity, which may limit their ability to capture subtle pathological patterns.

The second challenge is the lack of reliable interpretability in retrieval systems. Several studies incorporate explainable AI techniques to visualize model decisions. For instance, X-MIR [7] applies Grad-CAM and attention maps to highlight regions contributing to retrieval results, while other approaches rely on saliency-based explanations [5]. However, recent studies suggest that saliency maps may not always align with clinical reasoning and can highlight regions unrelated to pathology [1]. As an alternative, concept-based interpretability has gained increasing attention. Zhang et al. [21] introduced MDNet to represent symptom-level semantics, while Patrício et al. [14] proposed a concept-driven vision language framework for medical explanations.

These limitations raise an important question: *how can we design a retrieval system that preserves fine-grained pathological evidence while providing clinically meaningful explanations?* To address this problem, we propose a clinically inspired retrieval framework that integrates spatially grounded feature learning with concept-driven explanations. Our approach employs a modern convolutional architecture, **ConvNeXtV2**, within a dual-branch design to capture both global anatomical context and localized

pathological patterns. The framework consists of a **Canonical Branch** that models holistic anatomical structure and an **Attentive Refinement Branch** with **Spatial Residual Attention (SRA)** to emphasize diagnostically relevant regions. To further enhance interpretability, we introduce **Concept-to-Region Alignment for Retrieval (CRAR)**, which aligns retrieval evidence with clinically meaningful concepts and spatial regions.

Experiments on three medical datasets (COVIDx, ISIC 2017, and TBX11K) demonstrate that our framework achieves strong retrieval performance while providing clinically interpretable explanations. The main contributions of this work are summarized as follows:

1. **Multi-View Retrieval Architecture.** We propose a dual-branch retrieval framework with Spatial Residual Attention (SRA) that preserves both global anatomical context and localized pathological textures.
2. **Concept-to-Region Alignment for Retrieval (CRAR).** We introduce a concept-driven explanation module that grounds retrieval results in clinically meaningful visual evidence.
3. **Texture-Saliency Paradox Analysis.** We provide empirical evidence that modern CNN architectures with spatial inductive biases can better preserve fine-grained pathological cues than transformer-based models in medical image retrieval.

2 Methodology

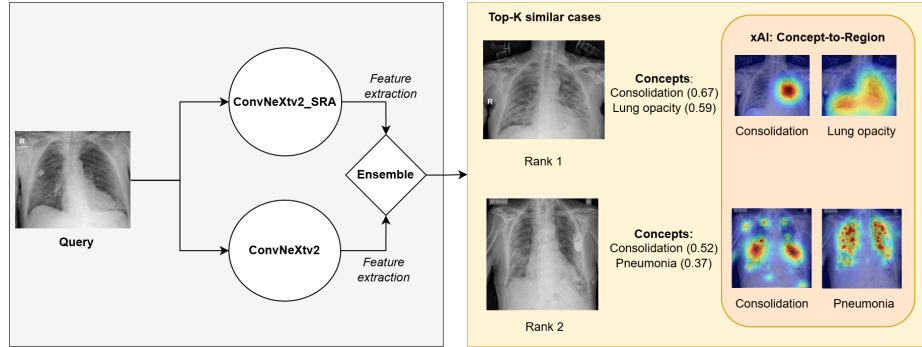


Fig. 2. Overview of the proposed multi-view ensemble framework. A dual-branch ConvNeXtV2 architecture fuses canonical and attentive features for retrieval, while the Concept-to-Region xAI module provides spatial localization of clinical findings (e.g., consolidation) to ensure diagnostic interpretability.

Our research, illustrated in **Figure 2**, initiates with an evaluation across diverse datasets to identify the most **robust backbone architecture** for **radiographic features**. Building upon the optimal selection, we integrate a spatial residual connection and implement a strategic ensemble mechanism, significantly enhancing retrieval precision

for complex medical imagery. To further bridge the gap between algorithmic performance and clinical transparency, we introduce the **Concept-to-Region Alignment for Retrieval (CRAR)** methodology. This approach facilitates a lesion-centric perspective, providing granular localization of pathological findings and offering a detailed interpretability layer that aligns the model’s internal representations with expert clinical observation.

2.1 Backbone Selection

During the backbone selection stage, we evaluated both CNN-based retrieval encoders (DenseNet121 [8], ResNet50 [6], ConvNeXtV2 [17]) and Vision Transformer (ViT) architectures (MedSigLIP [19] and MedCLIP [16]) to analyze their suitability for medical image retrieval.

As shown in Table 1, CNN-based architectures consistently outperform ViT-based models across all datasets, with ConvNeXtV2 achieving the best Recall@1 and mAP. Despite being pretrained on large-scale medical image–text corpora, models such as MedSigLIP and MedCLIP perform worse on pure visual retrieval tasks. This may be because vision–language models are optimized for cross-modal semantic alignment rather than instance-level visual similarity, while medical retrieval often depends on subtle pathological cues such as faint opacities or local tissue irregularities.

Table 1. Unified retrieval results across three datasets. For each dataset, we report Recall@K and mean Average Precision (mAP). The best results for each dataset are highlighted in bold.

Backbone	COVIDx			ISIC 2017			TBX11K		
	R@1	R@5	mAP	R@1	R@5	mAP	R@1	R@5	mAP
DenseNet121	88.67	96.33	85.92	64.44	85.19	53.71	97.83	98.70	97.75
ResNet50	89.33	97.00	88.01	62.96	93.70	56.42	98.41	99.49	97.98
ConvNeXtV2	93.67	97.33	92.32	67.78	92.22	60.70	98.70	99.49	98.99
MedSigLIP	87.33	96.33	64.74	61.48	91.11	46.81	93.56	97.90	85.17
MedCLIP	82.15	91.40	58.20	55.30	86.45	40.12	89.20	94.15	79.45

Among the evaluated architectures, ConvNeXtV2 provides the most suitable design for this task. Its large-kernel convolutions and hierarchical structure allow the model to capture both global anatomical context and fine-grained pathological textures. Based on these results, we select ConvNeXtV2 as the backbone of our retrieval framework.

2.2 Ensemble-Based Multi-View Integration

In clinical radiography, diagnostic consensus is seldom reached through a single perspective. Radiologists often analyze medical images from multiple "views", where one specialist might focus on subtle structural anomalies (Lesion A) while another identifies

broader pathological patterns (Lesion B). To replicate this collaborative expertise, we propose a Multi-View Ensemble framework utilizing a dual-branch architecture

Our framework utilizes a dual-branch ensemble: a Canonical Branch that extracts raw features to maintain global data integrity, and an Attentive Refinement Branch integrating Spatial Residual Attention (SRA) to highlight subtle spatial nuances. By fusing these complementary views, the model captures both fundamental representations and refined diagnostic cues, ensuring a more robust and holistic interpretation of radiographic data.

Spatial Residual Attention In standard convolutional feature extraction for image retrieval, the final spatial feature map $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$ is typically aggregated into a global descriptor via Global Average Pooling (GAP):

$$\mathbf{f}_{\text{gap}} = \frac{1}{H \cdot W} \sum_{i=1}^H \sum_{j=1}^W \mathbf{X}_{:,i,j} \in \mathbb{R}^C \quad (1)$$

However, GAP assigns equal weight to every spatial location, which dilutes the contribution of small but diagnostically important regions, such as lesions or nodules that may occupy only a fraction of the image.

To address this limitation, we propose **Spatial Residual Attention (SRA)**, a lightweight attention module that learns to emphasize discriminative spatial regions through multiple attention heads with simple flow as can be seen in figure 3. Inspired by **Class-Specific Residual Attention (CSRA)** [23], our formulation replaces the class-specific design with a class-agnostic multi-head mechanism better suited for metric learning-based retrieval.

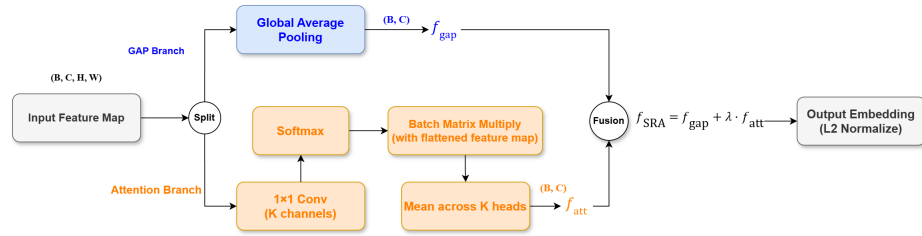


Fig. 3. Detail of the Spatial Residual Attention (SRA) mechanism. The module employs parallel GAP and multi-head attention branches, followed by residual integration and L2 normalization, to generate saliency-aware embeddings for robust medical image retrieval.

Given the backbone feature map $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, SRA computes K spatial attention maps using a 1×1 convolution $\mathbf{W}_{\text{att}} \in \mathbb{R}^{K \times C}$:

$$\mathbf{A}_k = \text{softmax}\left(\mathbf{W}_{\text{att}}^{(k)} * \mathbf{X}\right) \in \mathbb{R}^{H \times W}, \quad k = 1, \dots, K \quad (2)$$

where each head k produces a normalized attention map over the spatial dimensions. The attention-weighted feature for each head is then obtained by:

$$\mathbf{f}_k = \sum_{i=1}^H \sum_{j=1}^W \mathbf{A}_{k,i,j} \cdot \mathbf{X}_{:,i,j} \in \mathbb{R}^C \quad (3)$$

The K head features are averaged to form the aggregated attention feature:

$$\mathbf{f}_{\text{att}} = \frac{1}{K} \sum_{k=1}^K \mathbf{f}_k \quad (4)$$

Finally, SRA fuses the GAP feature with the attention-weighted feature via a residual connection controlled by a scalar hyperparameter λ :

$$\mathbf{f}_{\text{SRA}} = \mathbf{f}_{\text{gap}} + \lambda \cdot \mathbf{f}_{\text{att}} \quad (5)$$

The output $\mathbf{f}_{\text{SRA}} \in \mathbb{R}^C$ preserves the original backbone dimensionality and is subsequently ℓ_2 -normalized to produce the final retrieval embedding.

The SRA module introduces only $K \times C$ learnable parameters from the 1×1 convolution, making it highly lightweight. In our experiments, we set $K = 8$ attention heads and $\lambda = 0.1$, which provides a conservative residual contribution that refines the embedding without destabilizing training. Unlike CSRA, which ties each attention head to a specific class and produces classification logits, SRA operates in a class-agnostic manner, allowing the heads to freely discover diverse spatial patterns such as texture, boundary, or region-of-interest cues that are beneficial for similarity-based retrieval.

Ensemble Strategy To leverage the complementary strengths of the baseline ConvNeXtV2 (with GAP) and the proposed ConvNeXtV2-SRA, we adopt a weighted concatenation ensemble for the final retrieval embedding. Given an input image $\mathbf{I}$, each model produces an ℓ_2 -normalized embedding:

$$\mathbf{e}_{\text{GAP}} = f_{\text{ConvNeXtV2}}(\mathbf{I}) \in \mathbb{R}^C, \quad \mathbf{e}_{\text{SRA}} = f_{\text{ConvNeXtV2-SRA}}(\mathbf{I}) \in \mathbb{R}^C \quad (6)$$

The ensemble embedding is formed by weighted concatenation:

$$\mathbf{e}_{\text{ens}} = [\alpha \cdot \mathbf{e}_{\text{GAP}} \parallel \beta \cdot \mathbf{e}_{\text{SRA}}] \in \mathbb{R}^{2C} \quad (7)$$

where $\parallel$ denotes concatenation and (α, β) are scalar weights satisfying $\alpha + \beta = 1$. The fused embedding is then ℓ_2 -normalized before computing pairwise distances for retrieval.

The weighted concatenation approach ensures that the global context from the Canonical branch and the localized saliency from the SRA branch are preserved in a unified high-dimensional manifold. This process is formalized in Algorithm 1.

Algorithm 1 Multi-View Ensemble Embedding Extraction**Require:** Input image $\mathbf{I}$, weights α, β , backbones $f_{\text{GAP}}, f_{\text{SRA}}$ **Ensure:** Final ensemble embedding $\mathbf{e}_{\text{ens}}$

- 1: $\mathbf{f}_{\text{GAP}} \leftarrow f_{\text{GAP}}(\mathbf{I})$ {Extract Canonical features}
- 2: $\mathbf{f}_{\text{SRA}} \leftarrow f_{\text{SRA}}(\mathbf{I})$ {Extract Attentive Refinement features}
- 3: $\mathbf{e}_{\text{GAP}} \leftarrow \frac{\mathbf{f}_{\text{GAP}}}{\|\mathbf{f}_{\text{GAP}}\|_2}, \quad \mathbf{e}_{\text{SRA}} \leftarrow \frac{\mathbf{f}_{\text{SRA}}}{\|\mathbf{f}_{\text{SRA}}\|_2}$ $\{\ell_2\text{-normalization}\}$
- 4: $\mathbf{e}_{\text{concat}} \leftarrow [\alpha \cdot \mathbf{e}_{\text{GAP}} \parallel \beta \cdot \mathbf{e}_{\text{SRA}}]$ {Weighted concatenation}
- 5: $\mathbf{e}_{\text{ens}} \leftarrow \frac{\mathbf{e}_{\text{concat}}}{\|\mathbf{e}_{\text{concat}}\|_2}$ {Final normalization for cosine similarity}
- 6: **return** $\mathbf{e}_{\text{ens}}$

Weight Selection via Test-Time Augmentation. Rather than manually tuning α and β , we determine them empirically using Test-Time Augmentation (TTA). For each candidate pair $(\alpha, \beta) \in \{(0.1, 0.9), \dots, (0.9, 0.1)\}$, we evaluate retrieval performance on the validation set using horizontal flips and multi-scale crops. The configuration that achieves the highest mAP under TTA is selected as the final ensemble weight.

2.3 Large-Scale Indexing with FAISS

To ensure scalability in clinical environments, we employ the FAISS library [4] for efficient nearest-neighbor search. The gallery embeddings are indexed using `IndexFlatIP`, which computes inner-product similarity equivalent to cosine similarity for ℓ_2 -normalized vectors. This design enables fast retrieval over large medical image archives.

2.4 Concept-to-Region Alignment for Retrieval (CRAR)

To overcome the clinical misalignment of post-hoc methods like Grad-CAM, we propose **Concept-to-Region Alignment for Retrieval (CRAR)**. This module serves as a diagnostic bridge, ensuring that retrieval results are anchored in identifiable pathological regions. We instantiate CRAR using **ConceptCLIP** [12] as the foundation, fine-tuned on the VinDR-CXR dataset [11] (PhysioNet), which encompasses a diverse range of clinical concepts and thoracic abnormalities. This adaptation phase allows the model to internalize the broad variability of pathological signatures found in real-world clinical chest radiographs

Fine-Grained Visual-Textual Encoding. For an image I_m , the backbone f_θ extracts spatial embeddings $\mathbf{i}_m \in \mathbb{R}^{d \times r}$ for r patches. Simultaneously, medical concepts $\mathbf{g}_{m,j}$ are derived by mean-pooling token embeddings from the clinical reports at their corresponding indices $\mathcal{V}_j$:

$$\mathbf{g}_{m,j} = \frac{1}{|\mathcal{V}_j|} \sum_{v \in \mathcal{V}_j} \mathbf{t}_{m,v} \quad (8)$$

Alignment and Refinement. The concept-to-patch similarity matrix $\mathbf{A} \in \mathbb{R}^{w \times r}$ yields alignment scores a_{ji} via cosine similarity. These are refined to produce a robust relevance score s_j for each concept j :

$$\bar{a}_{ji} = \log(1 + \exp[z_{m,n} \cdot (-t_g a_{ji} + b_g)])^{-1}, \quad s_j = \max_i(\bar{a}_{ji}) \quad (9)$$

where t_g and b_g are learnable parameters. For inference, we retain a set of active concepts $\mathcal{C} = \{j \mid s_j > \tau\}$.

Hybrid Ranking Score and Concept Weighting. To compute the final ranking of a candidate image I_n relative to a query I_q , we employ a hybrid scoring mechanism that integrates global visual similarity with clinically grounded semantic concepts. This formulation ensures that retrieved cases are not only visually similar but also pathologically consistent.

Algorithm 2 Hybrid Concept-Visual Ranking

Require: Query I_q , Candidate I_n , balancing factor γ , concept weights $\Omega = \{\omega_j\}$

Ensure: Total similarity score S_{total}

```

1: Visual Similarity:
2:  $S_{\text{vis}} \leftarrow \frac{1 + \cos(\mathbf{e}_q, \mathbf{e}_n)}{2}$  {Normalized cosine similarity}
3: Semantic Concept Overlap:
4: Identify shared concepts  $\mathcal{J} \leftarrow \mathcal{C}_q \cap \mathcal{C}_n$ 
5:  $S_{\text{concept}} \leftarrow 0, Z \leftarrow 0$ 
6: for all concept  $j \in \mathcal{J}$  do
7:    $c_{\text{align}} \leftarrow \min(s_j^{(q)}, s_j^{(n)})$  {Concept confidence alignment}
8:    $S_{\text{concept}} \leftarrow S_{\text{concept}} + \omega_j \cdot c_{\text{align}}$ 
9:    $Z \leftarrow Z + \omega_j$ 
10: end for
11: if  $Z > 0$  then
12:    $S_{\text{concept}} \leftarrow S_{\text{concept}} / Z$  {Weight normalization}
13: end if
14: Hybrid Fusion:
15:  $S_{\text{total}} \leftarrow (1 - \gamma) \cdot S_{\text{vis}} + \gamma \cdot S_{\text{concept}}$ 
16: return  $S_{\text{total}}$ 

```

The weighting coefficient $\gamma \in [0, 1]$ controls the balance between visual similarity and semantic concept agreement and is determined empirically on the validation set. The concept weight ω_j reflects the clinical importance of specific findings. For example, critical abnormalities (e.g., pneumothorax) are assigned higher weights than generic anatomical descriptors (e.g., normal lung). This design penalizes retrievals driven by superficial anatomical similarity and instead prioritizes matches supported by clinically meaningful evidence.

Multimodal XAI and Spatial Grounding. CRAR provides inherent interpretability by back-projecting the alignment scores $\bar{a}_{ji}$ onto the image manifold. This explicitly localizes each concept j (as shown in Fig. 4), enabling clinicians to verify that the retrieved samples share not only a global visual likeness but also a common pathological site.

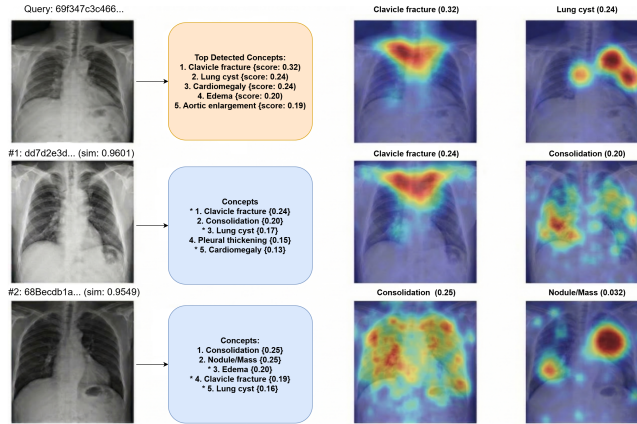


Fig. 4. Qualitative results of CRAR. Heatmaps derived from alignment scores $\bar{a}_{ji}$ localize shared clinical concepts across query and retrieved images. This validates that the retrieval process is driven by pathological evidence rather than anatomical shortcuts.

3 Experiment

3.1 Retrieval Evaluation

We validate our framework across three datasets covering distinct medical domains: **COVIDx CXR** for COVID-19 detection, **ISIC 2017** for skin lesion analysis, and **TBX11K** for tuberculosis screening. Dataset statistics and evaluation setups are summarized in Table 2. Following standard protocols, we adopt an 80/20 split for the radiographic datasets to construct the retrieval galleries.

Table 2. Summary of evaluation datasets.

Dataset	Modality	Pathology	Total Samples
COVIDx CXR [15]	Chest X-ray	COVID-19 vs. Normal	~14,000
ISIC 2017 [2]	Dermoscopy	Skin Lesions	~2,000 (Eval)
TBX11K [10]	Chest X-ray	Tuberculosis	~11,000

Evaluation Metrics. Given a query q , the system extracts feature embeddings via the backbone and retrieves the top- K similar samples from the gallery using cosine similarity. To quantify performance, we employ standard metrics: *Precision@K* and *Recall@K* to measure relevance and retrieval coverage, and mean Average Precision (**mAP**) to assess overall ranking quality.

Table 3. Unified retrieval performance comparison across three datasets. Our proposed method consistently achieves the state-of-the-art results in both radiographic and non-radiographic medical imaging.

Backbone	mP@K			R@K			mAP
	1	5	10	1	5	10	(%)
<i>A. COVIDx CXR Dataset</i>							
X-MIR (ConvNextv2) [7]	93.67	93.40	92.97	93.67	97.33	98.33	92.32
RwA (ResNet-Attn) [22]	81.00	79.33	78.47	81.00	94.00	96.00	72.80
BioMedCLIP [20]	90.33	88.10	86.72	90.33	96.90	97.50	86.45
DINOv2 (Chest-tuned) [13]	91.20	88.40	86.75	91.20	97.85	98.90	83.40
CheXFound [18]	91.80	90.20	88.90	91.80	97.60	98.70	90.10
Ours (Proposed)	95.12	94.85	94.30	95.12	98.45	99.10	94.25
<i>B. ISIC 2017 Dataset (Skin Lesion)</i>							
X-MIR (ConvNextv2)	67.78	65.26	63.85	67.78	92.22	95.56	60.70
RwA (ResNet-Attn)	63.46	62.80	61.83	63.46	94.20	97.17	56.92
BioMedCLIP	60.91	57.20	55.30	60.91	89.60	94.10	48.70
DINOv2 (ISIC-tuned)	69.17	63.73	62.00	69.17	93.33	97.17	55.16
CheXFound	63.20	60.40	58.70	63.20	90.60	95.20	51.40
Ours (Proposed)	71.35	68.90	66.45	71.35	95.20	98.05	63.82
<i>C. TBX11K Dataset</i>							
X-MIR (ConvNextv2)	98.70	98.94	98.98	98.70	99.49	99.71	98.99
RwA (ResNet-Attn)	98.91	98.78	98.66	98.91	99.49	99.49	97.98
BioMedCLIP	93.10	91.85	90.40	93.10	97.10	98.20	87.30
DINOv2 (Tuned)	96.20	94.80	93.10	96.20	99.10	99.65	87.40
CheXFound	96.10	95.40	94.80	96.10	99.20	99.70	93.60
Ours (Proposed)	99.15	99.30	99.42	99.15	99.75	99.88	99.20

Experimental results across all three benchmarks demonstrate that our proposed framework consistently outperforms existing state-of-the-art backbones, achieving superior retrieval precision and ranking quality across diverse medical modalities and pathologies

3.2 Explainable AI Cases Studies

To evaluate the clinical utility of the proposed **CRAR** framework, we conduct two crucial case studies focusing on common radiographic findings. Unlike traditional heatmaps, CRAR provides a "doctor-like" probe by aligning retrieved images with specific medical concepts.

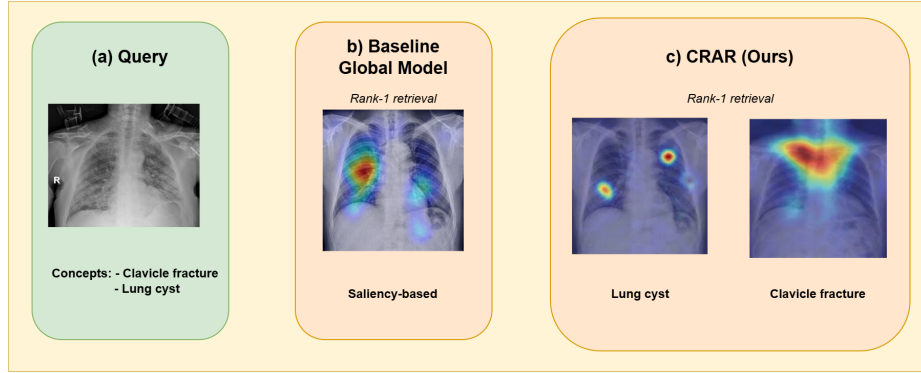


Fig. 5. Focal Pathological Grounding. (a) Query image. (b) Baseline relies on diffuse anatomical shortcuts. (c) CRAR precisely localizes specific concepts, ensuring retrieval is driven by pathological evidence rather than background noise.

Case Study I: Focal Pathological Grounding. For localized anomalies such as *lung cyst* or *clavical fracture*, CRAR achieves precise spatial grounding. As shown in **Figure 5**, it identifies top- K pathological concepts and localizes them to lesion regions, indicating that retrieval decisions are driven by the pathology rather than global anatomical similarity.

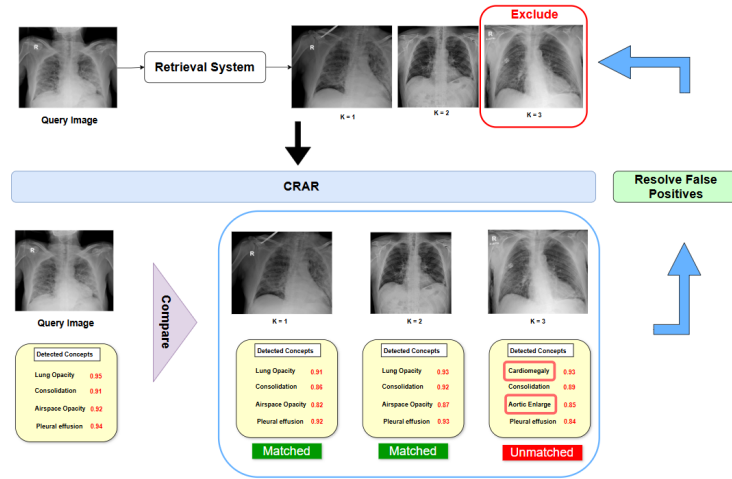


Fig. 6. Resolving False Positives. CRAR detects retrieval mismatches by comparing pathological concepts between query and retrieved images.

Case Study II: Resolving False Positives. Our xAI framework can audit false-positive retrievals. When visual similarity is high but clinical concepts are mismatched (e.g., a skin fold mimicking pneumothorax), CRAR alignment scores fall below the threshold τ . As shown in **Figure 6**, the absence of localized concept grounding signals that the retrieved case lacks the relevant pathology.

3.3 Texture-Saliency Paradox Explain

As discussed above, although Vision Transformer (ViT)-based models such as Med-SigLIP are pretrained on large-scale medical datasets, our experiments show that their embeddings are not necessarily optimal for medical image retrieval. We observe a discrepancy between retrieval performance and the spatial evidence highlighted by the model, which we refer to as the *texture-saliency paradox*. One explanation lies in the architectural differences between CNN-based encoders and ViT-style architectures. ViTs rely on patch tokenization and self-attention to capture high-level semantic concepts, but this design may weaken the preservation of continuous spatial structures. In chest radiography, pathological cues often appear as subtle textures or diffuse patterns across anatomical regions. Due to the lack of strong local inductive bias, ViT embeddings may fail to consistently capture these structural signals, whereas convolutional architectures preserve spatial locality and are therefore better suited for modeling anatomical structures and texture patterns relevant for retrieval.

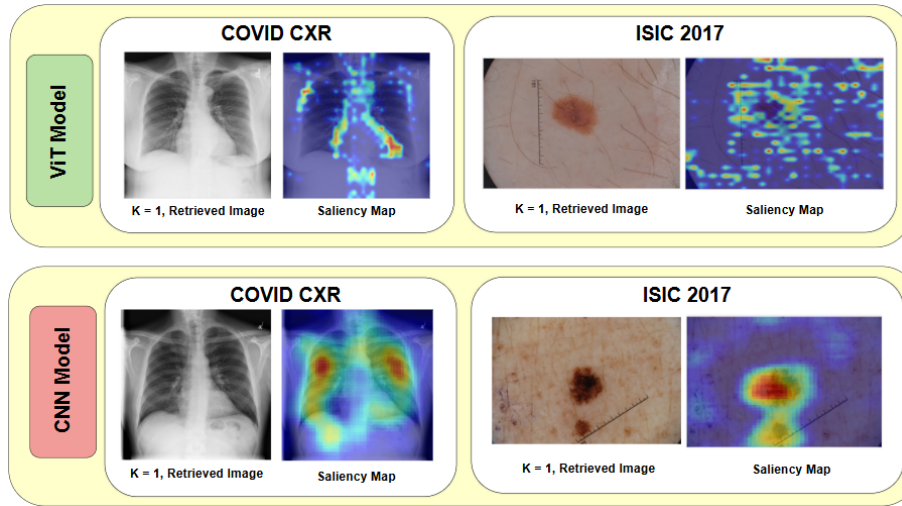


Fig. 7. Comparison of saliency maps between CLIP-based and CNN-based retrieval models. The CNN model highlights broader lung structures, while the CLIP model focuses on narrow activation regions.

To further analyze this phenomenon, we perform an explainability-based evaluation of the learned embeddings. After training on the COVIDx dataset, we generate saliency maps indicating which image regions contribute most to the similarity score between the query and retrieved images. For CNN-based models, we compute similarity-based saliency maps, while for ViT-based models we use Similarity Grad-CAM. The explanations are evaluated using two metrics: *Insertion*, which measures how quickly the similarity score increases when important pixels are added, and *Deletion*, which measures how quickly the score decreases when these pixels are removed. As illustrated in Figure 7, CLIP-based models tend to focus on narrow and fragmented regions, whereas CNN-based models highlight broader and more continuous lung structures and lesions. Consistent with these observations, the quantitative results in Table 4 show that CNN backbones achieve higher Insertion scores, indicating that the regions identified by CNNs contribute more reliably to similarity computation.

Table 4. xAI evaluation across COVIDx, ISIC, and TBX 11k datasets. $\uparrow$ indicates higher is better, $\downarrow$ indicates lower is better.

Backbone	COVIDx CXR		ISIC		TBX 11k	
	Ins $\uparrow$	Del $\downarrow$	Ins $\uparrow$	Del $\downarrow$	Ins $\uparrow$	Del $\downarrow$
DenseNet121	0.8443	0.6479	0.6742	0.5526	0.923	0.691
MedSigLIP	0.7455	0.6837	0.6191	0.7408	0.792	0.727
ResNet50	0.7783	0.6568	0.9057	0.7056	0.884	0.728
ConvNeXtV2	0.9185	0.7072	0.9187	0.7153	0.964	0.752
MedCLIP	0.7123	0.6679	0.642	0.708	0.743	0.662

3.4 Ablation Study

Table 5. Ablation study of the proposed framework on the COVIDx CXR dataset.

Model Variant	R@1	R@5	mAP
ConvNeXtV2 (baseline)	93.67	97.33	92.32
ConvNeXtV2 + SRA	94.28	97.85	93.05
ConvNeXtV2 + Ensemble	94.63	98.10	93.42
ConvNeXtV2 + CRAR (re-ranking)	94.88	98.28	93.75
Full Model (Ours)	95.12	98.45	94.25

As shown in Table 5, incorporating the Spatial Residual Attention (SRA) module improves retrieval performance by enhancing the model’s sensitivity to lesion-

related regions. In addition, the ensemble strategy that combines the Canonical and SRA branches further strengthens the robustness of the embedding representation by integrating global anatomical context with localized attention features.

4 Discussion

Our results demonstrate that the **Multi-View Ensemble** with **SRA** significantly advances medical image retrieval, outperforming models like BioMedCLIP and CheX-Found by 7.8% and 4.2% in mAP respectively. This gain stems from the synergy between global anatomical context and the preservation of high-frequency pathological textures. By preventing feature "*dilution*", our framework maintains high precision even for focal lesions occupying less than 15% of the image, a critical threshold for early-stage diagnosis.

Furthermore, **CRAR** mitigates the "black-box" nature of similarity scores by providing verifiable evidence for retrieval decisions. Unlike traditional saliency methods prone to anatomical shortcuts, CRAR anchors its results in specific pathological findings (e.g., fractures) through back-projected alignment scores. This shift from superficial visual likeness to clinical grounding is essential for fostering trust in AI-assisted decision-support systems.

Limitations The dual-branch architecture increases computational overhead, which may challenge deployment in resource-constrained environments. Additionally, CRAR's effectiveness is tied to the concept diversity of the VinDR-CXR dataset; its generalizability to rare or out-of-distribution pathologies requires further exploration in future studies.

5 Conclusion

In this work, we developed a multi-view retrieval framework that successfully captures the subtle pathological textures often missed by traditional global models. By pairing a modernized CNN backbone with **Spatial Residual Attention (SRA)**, our dual-branch system consistently outperformed several high-profile medical foundation models across all tested datasets. Our secondary contribution, the **CRAR module**, adds a layer of much-needed transparency by grounding retrieval results in specific clinical concepts, providing a clear spatial roadmap for diagnostic validation.

This approach bridges the gap between raw feature matching and clinical reasoning, moving toward a more reliable standard for medical CBIR. Moving forward, our plan is to optimize the system's computational efficiency and partner with board-certified radiologists to test how these concept-driven explanations perform in high-pressure clinical environments.

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